

Measurement-induced phase transition in random hybrid Clifford circuits: exact annealed criticality and a high-precision quenched locator

Abstract

We study the measurement-induced entanglement transition in random hybrid circuits of L qudits (local dimension d) on a ring: two layers of random two-site Clifford gates per period, each followed by independent computational-basis measurements at rate p . For the *annealed* (disorder-averaged) two-replica problem the full moment dynamics closes on a three-dimensional per-site commutative algebra and is exactly similar to the row transfer of an anisotropic triangular-lattice Ising model, $\mathcal{C} = \mathcal{W}_0^L E D_h E^\top D_h$ acting on $2^{L/2}$ states, with $K_d = \frac{1}{4} \ln(a/c)$, $K_h = \frac{1}{4} \ln(ac/b^2)$, $\mathcal{W}_0 = (acb^2)^{1/4}$ determined by the measurement channel. We prove a closed form for the complete nonzero spectrum (a free-fermion product over parity-quantised momenta, recovering the rank $3 \cdot 2^{L/2-2}$ by mode counting), and obtain the exact annealed critical line $p_c^{(2)}(d) = (d - \vartheta_c)/(d - 1)$, $\vartheta_c = \beta_d + \sqrt{\beta_d^2 + 1}$, $\beta_d = (d^2 - 1)/(d^2 + 1)$, with certified leading eigenvalues (exact-rational Collatz–Wielandt enclosures of relative width $\leq 7.3 \times 10^{-15}$). The record-collision weight of the annealed channel favours trajectories with fewer measurements, lowering the effective monitoring density to $p_{\text{eff}} = p + \frac{p(1-p)}{2L} \partial_p \ln \lambda_1$; this resolves the annealed-versus-quenched entropy ordering without a convexity argument. For the *quenched* Clifford dynamics we locate the transition from 475,600 trajectories ($L = 16$ –512) of the tripartite information I_3 and the logarithmic entanglement entropy: crossings, windowed data collapse, bootstrap errors): $p_c = 0.1597(8)$, $\nu = 1.24(7)$, and critical half-chain entropy $\alpha = 1.55(7)$ bits per $\ln L$ (systematic rather than statistical uncertainties). An independent purification locator (Gullans–Huse reference-qubit protocol, disjoint seeds, mixed initial state, 135,000 trajectories) gives $p_c^{\text{purif}} = 0.1601$ –0.1604, $\nu^{\text{purif}} = 1.25(2)$. Our values agree with Gullans–Huse 0.1593(5), $\nu = 1.28(2)$ and Sierant et al. 0.15995(10). Finally, the record statistics admit a quenched scaled cumulant generating function whose $\beta = 1$ endpoint is exactly the two-replica partition function: we verify its thermodynamic limit (the $\beta \rightarrow 1$ side converging to the Houtappel free energy per site, the quenched information density to 0.1489 bits per measurement site with an L -independent per-site variance), demonstrate the freezing of the collision tilt (carried by $O(1)$ – $O(10)$ records at $L = 8$ –24), and show that the annealed-minus-quenched free-energy gap per site, 0.0222(5) nats, is L -independent: the record measure is a glass whose observables nevertheless self-average.

1 Introduction

Hybrid random circuits—interleaving unitary evolution with projective measurements—display a measurement-induced entanglement transition between a volume-law and an area-law phase (Skinner et al., 2019; Li et al., 2018; Gullans and Huse, 2020; Nahum et al., 2020). Numerically the transition is located through disorder-averaged observables (I_3 , entropies, purification) whose sample-to-sample fluctuations grow exponentially with system size; in monitored *Clifford* circuits those averages can be sampled exactly by stabilizer simulation, which makes the Clifford ensemble

the natural high-precision testbed (Gullans and Huse, 2020; Sierant et al., 2022; Zabalo et al., 2020).

This paper combines two complementary attacks on the same Clifford model. (i) *Exact annealed results*. The second moment of the monitored channel averaged over circuits (equivalently over measurement *records* with the record-collision weight $P(R)^2$) is exactly solvable: the two-replica dynamics closes on a per-site algebra of three commuting projectors, compresses to a classical transfer matrix of Ising type, and yields the full finite-size spectrum in closed form, the exact annealed critical line for every local dimension d , and certified eigenvalues. (ii) *Quenched numerics*. A high-precision stabilizer simulation of the *Born-sampled* (quenched) dynamics pins the transition at $p_c = 0.1597(8)$ with $\nu = 1.24(7)$ from 475,600 trajectories, and an independent Gullans–Huse purification protocol with disjoint seeds confirms the point through a different observable.

The paper is organised as follows. Sec. 2 defines the circuit and the two-replica (record-collision) moment problem, introduces the per-site algebra and the compressed classical transfer (Eqs. 2–5), and states the rank theorem. Sec. 3 identifies the classical model as the anisotropic triangular-lattice Ising model, recalls the Houtappel free energy (Prop. 1), and proves the closed-form spectrum (Prop. 2) together with its numerical certificate. Sec. 4 verifies the exact solution against the finite-size benchmark (Table 1) and the certified eigenvalues (App. A.4). Sec. 5 derives the exact annealed critical line and the effective-density (tilt) mechanism. Sec. 6 states the multifractal record statistics. Sec. 7 closes the record-level theorem items: the quenched record SCGF and its thermodynamic limit, the freezing of the collision tilt, the disorder–replica interchange, and the complexity status of the Born averages. Sec. 8 presents the quenched locator and the adopted numbers; Sec. 9 concludes. App. A collects the rank certificates and the exact-rational enclosures; App. B documents the numerical estimators, their error model, the audit-driven robustness analysis, and the independent purification locator; App. C documents the exact two-replica evolution and the record-SCGF verification suite.

2 Model, two-replica compression, and the rank theorem

2.1 The monitored Clifford circuit

The system is a ring of $L = 2m$ qudits (dimension d , computational basis $\{|x\rangle\}_{x=0}^{d-1}$). One circuit *period* consists of two brickwork layers of random two-qudit gates (even bonds $(2k, 2k+1)$, then odd bonds $(2k+1, 2k+2)$, each gate drawn independently and uniformly from the two-qudit Clifford group), each gate layer followed by a measurement layer in which every site is independently measured in the computational basis with probability p . A *record* R is the complete list of measurement positions and outcomes along a trajectory; its Born probability under a random circuit realisation is $P(R)$. The quenched observables of interest (I_3 , half-chain entropy $S_{L/2}$) are Born-sampled averages over trajectories; the annealed second moments average the record probability squared, which glues two replicas onto the same record.

2.2 The record-collision moment and the per-site algebra

The two-replica state σ lives on $(\mathbb{C}^d)^{\otimes L} \otimes (\mathbb{C}^d)^{\otimes L}$. Under a gate U acting identically on both replicas, $\sigma \mapsto (U \otimes U) \sigma (U^\dagger \otimes U^\dagger)$; since the two-qudit Clifford group is an exact 2-design, the circuit average of this conjugation is the twirl onto the commutant $\text{span}\{I, S\}$ of each gate bond (S the replica swap). A record-colliding measurement of site i maps $\sigma \mapsto (\Pi_i \otimes \Pi_i) \sigma (\Pi_i \otimes \Pi_i)$

with $\Pi_i = \sum_x |x\rangle\langle x|$ the same-outcome projector on both replicas. The relevant operator algebra is therefore generated per site by

$$e_0 = \Pi, \quad e_+ = \frac{I+S}{2} - \Pi, \quad e_- = \frac{I-S}{2}, \quad (1)$$

three mutually orthogonal projectors ($e_0 + e_+ + e_- = I$; $Se_0 = e_0$, $Se_+ = e_+$, $Se_- = -e_-$). The gate twirl preserves this algebra and the colliding measurement acts diagonally on it, $e_0 \mapsto e_0$, $e_\pm \mapsto (1-p)e_\pm$; the whole annealed dynamics is a positive linear map on the 3^L -dimensional commutative algebra spanned by the products $\bigotimes_k e_{a_k}$. Because the transfer is similar (Schur complement on the collided sector) to a 2^m -dimensional classical transfer acting on the pair variables, all annealed $n = 2$ quantities coincide with the Haar ones by the 2-design property (Remark ??).

2.3 The compressed classical transfer

Let $s = (s_1, \dots, s_m) \in \{\pm 1\}^m$ label the pair (row) configuration. The half-period transfer is the $2^m \times 2^m$ matrix

$$M_1 = \mathcal{W}_0^m E D_h, \quad E_{s's} = \exp\left[K_d \sum_k s'_k (s_k + s_{k+1})\right], \quad D_h = \text{diag} \exp\left[K_h \sum_k s_k s_{k+1}\right], \quad (2)$$

with $s_{m+1} = s_1$ (ring). The full period map is

$$\mathcal{C} = M_1 M_2 = \mathcal{W}_0^L E D_h E^\top D_h = M_1^2 \tau, \quad (3)$$

where $M_2 = \mathcal{W}_0^m E^\top D_h = M_1 \tau$ and τ is the bond-ring shift ($(E^\top = E\tau, [D_h, \tau] = 0)$). The bond weight of a single pair transition factorises as $w(s'|s, s^+)$ with

$$a = w(+|+, +), \quad b = w(\pm|+, -), \quad c = w(-|+, +), \quad (4)$$

and the Ising couplings are (Eq. 5)

$$K_d = \frac{1}{4} \ln \frac{a}{c}, \quad K_h = \frac{1}{4} \ln \frac{ac}{b^2}, \quad \mathcal{W}_0 = (acb^2)^{1/4}. \quad (5)$$

Writing $\vartheta = d - (d-1)p$, the measurement channel gives the weights in closed form, with two branches joined continuously at the annealed critical point $\vartheta_c = \beta_d + \sqrt{\beta_d^2 + 1}$, $\beta_d = (d^2 - 1)/(d^2 + 1)$:

$$\text{ordered } (\vartheta \geq \vartheta_c): \quad a = \frac{(\vartheta + 1)^2}{2(d^2 + 1)}, \quad b = \frac{\vartheta^2 - 1}{2(d^2 - 1)}, \quad c = \frac{(\vartheta - 1)^2}{2(d^2 + 1)}, \quad \mathcal{W}_0 = \frac{\vartheta^2 - 1}{2\sqrt{d^4 - 1}}, \quad (6)$$

$$\text{disordered } (\vartheta \leq \vartheta_c): \quad a = \frac{d^2 \vartheta^2 - 1}{d^4 - 1}, \quad b = \frac{\vartheta}{d^2 + 1}, \quad c = \frac{d^2 - \vartheta^2}{d^4 - 1}, \quad \mathcal{W}_0 = \frac{[\vartheta(d^2 - 1)]^{1/2} [(d^2 \vartheta^2 - 1)(d^2 - 1)]}{d^4 - 1} \quad (7)$$

At $\vartheta = \vartheta_c$ both branches give $a - c = 2b$ exactly: the $k = 0$ mode of Prop. 2 is gapless there. In the limits $p \rightarrow 1$ ($\vartheta \rightarrow 1$) the three weights coincide, $a = b = c = 1/(d^2 + 1)$ (the disordered fixed point), and at $p \rightarrow 0$ ($\vartheta \rightarrow d$) the ordered weights $a = (d+1)^2/2(d^2 + 1)$, $b = \frac{1}{2}$, $c = (d-1)^2/2(d^2 + 1)$ describe the strong-coupling (volume-law) limit.

Theorem 1 (rank; compressed period map). *Let $L = 2m \geq 4$, $d \geq 2$, $0 < p < 1$.*

(i) $\mathcal{C}$ is similar to a positive-semidefinite matrix $\mathcal{W}_0^L D_h^{1/2} Y Y^\top D_h^{1/2}$ with $Y = D_h^{-1/2} E D_h^{1/2}$; in particular its spectrum is real and non-negative.

(ii) $\mathcal{C}$ commutes with the global spin flip $F : (s \mapsto -s)$ and with the ring shift τ .

(iii) The nonzero spectrum splits into a flip-even and a flip-odd sector of 2^{m-1} states each.

(iv) $\text{rank } \mathcal{C} = 3 \cdot 2^{m-2}$ and $\dim \ker \mathcal{C} = 2^{m-2}$.

Proof sketch. (i) is the similarity $D_h^{-1/2} \mathcal{C} D_h^{1/2} = \mathcal{W}_0^L (D_h^{-1/2} E D_h^{1/2}) (D_h^{-1/2} E D_h^{1/2})^\top$. (ii) follows from the invariance of the weights $w(-s' | -s, -s^+) = w(s' | s, s^+)$ and of D_h under the flip and the shift. (iv) is proved mode-by-mode in Prop. 2(iii) below (the $k = \pi$ mode has $w_\pi^- = 0$); an independent Schur-complement proof through $L = 12$ is given in App. A. $\square$

3 The Ising form, the Houtappel limit, and the closed-form spectrum

E is the diagonal-bond (vertical) transfer of an anisotropic triangular-lattice Ising model with horizontal coupling K_h and the two diagonal couplings K_d ; $M_1 = E D_h$ is its row transfer. The finite-size scaling variable is the Ising reduced coupling; the annealed transition is the triangular-lattice critical line

$$\sinh(2K_d) e^{2K_h} = 1, \quad (8)$$

which at the physical trajectory (6)–(7) is met exactly at $\vartheta = \vartheta_c$, i.e. at $p = p_c^{(2)}(d) = (d - \vartheta_c)/(d - 1)$: $e^{2K_h} = \beta_d$ and $\sinh(2K_d) = 1/\beta_d$ there. (For the isotropic point $K_d = K_h$ Eq. (8) gives $\sinh(2K) e^{2K} = 1$, i.e. $K = \frac{1}{4} \ln 3$, the known triangular-lattice critical coupling.)

Proposition 1 (Houtappel). *The thermodynamic leading eigenvalue obeys $\frac{1}{L} \ln \lambda_1 \rightarrow \ln \mathcal{W}_0 + f_H(K_d, K_h)$ with f_H the anisotropic triangular-lattice free energy per site of Houtappel (1950). At $d = 2$, $p = 0.40$: $\lambda_1(L)^{1/L} \rightarrow 0.683844659$.*

Proposition 2 (closed form of the nonzero spectrum). *Let $L = 2m \geq 4$, $d \geq 2$, $0 < p < 1$, and let $\mathcal{C} = \mathcal{W}_0^L E D_h E^\top D_h$ be the operator of Eq. (3), with $K_d, K_h, \mathcal{W}_0$ as in Eq. (5). For $k \in [0, 2\pi)$ define*

$$Q(k) = \cosh^2 2K_d \cosh 2K_h + \sinh^2 2K_d \sinh 2K_h - \sinh 2K_h \cos k, \quad R(k) = 2 \sinh 2K_d \cos \frac{k}{2}, \quad (9)$$

and $w_k^\pm = Q(k) \pm \sqrt{Q(k)^2 - R(k)^2}$ (real, since $0 \leq R^2 \leq Q^2$ for all k). Then every nonzero eigenvalue of $\mathcal{C}$ is of the form

$$\lambda = (2\mathcal{W}_0^2)^m \prod_{k \in \mathcal{K}_\sigma} w_k^{\pm k}, \quad (10)$$

where the product runs over the momentum states of the flip sector σ : in the flip-even sector $\mathcal{K}_+ = \{(2j - 1)\pi/m : j = 1, \dots, m\}$ (antiperiodic) with an even number of “−” choices; in the flip-odd sector $\mathcal{K}_- = \{2\pi j/m : j = 0, \dots, m - 1\}$ (periodic) with an odd number of “−” choices when the branches are ordered as $w^\pm = Q \pm \sqrt{Q^2 - R^2}$ fixed. Equivalently, in the convention that orders the $k = 0$ branches by magnitude (the ordering flips at $p_c^{(2)}$), the flip-odd sector carries an even number of “−” choices below $p_c^{(2)}(d)$ and an odd number above, exactly as in the spinor analysis of Kaufman (1949); Newell (1950). Moreover:

- (i) $R(\pi) = 0$, so $w_\pi^- = Q(\pi) - |Q(\pi)| = 0$: every state occupying the “ $-$ ” branch of the $k = \pi$ mode lies in the kernel. Counting nonzero states (half of the sector containing $k = \pi$, all of the other) gives $\text{rank } \mathcal{C} = 3 \cdot 2^{m-2}$, recovering Theorem 1(iv) by mode counting.
- (ii) At $p = p_c^{(2)}(d)$ the $k = 0$ mode is gapless: $Q(0) - R(0)$ has a double zero at $a - c = 2b$, so $w_0^+ = w_0^-$ there, and the scaled gaps have the exact limits $L \ln(\lambda_1/\lambda_{\text{odd}}) \rightarrow 2\pi \beta_d/4$ and $L \ln(\lambda_1/\lambda_\varepsilon) \rightarrow 2\pi \cdot 2\beta_d$, ratio $1/8 = x_\sigma/x_\varepsilon$ for every d .
- (iii) $a - c = 2b \iff Q(0) = R(0)$; the critical line Eq. (8) is the gapless condition of the $k = 0$ mode.

Proof. The transfer E is a bond-factorised matrix whose spinor (Jordan–Wigner) analysis follows Kaufman (1949); Newell (1950); Stephenson (1964); Schultz et al. (1964): (a) in the spinor representation with open chain ends the diagonal transfer is a product of Ising–Clifford exponentials. With $x_k = \sigma_k^x$ and $z_k = \sigma_k^z$ (the Schultz–Mattis–Lieb algebra of the row transfer), K_1 the Kramers–Wannier dual of K_d ($\sinh 2K_1 \sinh 2K_d = 1$), $K_2 = K_d$ and $c = \sqrt{2 \sinh 2K_1}$,

$$E = \sinh^m(2K_d) [\cosh K_d \tilde{Y} + \sinh K_d z_m \tilde{Y} z_1], \quad \tilde{Y} = c e^{K_1 x_m} U_{m-1} \cdots U_1, \quad U_k = e^{K_d z_k z_{k+1}} c e^{K_1 x_k}, \quad (11)$$

the periodic boundary contributing only through the end term $z_m \tilde{Y} z_1$; every factor of (11) has unit determinant, so $\det \tilde{Y} = c^{m2^m}$, one flip sector of E is exactly singular (the $k = \pi$ kernel of item (i) below, at the transfer level), and the two flip sectors of the open-boundary transfer have equal determinants $|\det E_{\text{OBC}, \varepsilon}| = (2 \sinh 2K_d)^{m2^{m-2}}$. Identity (11) is machine-verified (relative residual $\leq 6 \times 10^{-16}$ for $m = 3-8$, $d \in \{2, 3, 5\}$, both branches of the weight trajectory; `spinor_verify.py`, checks V1–V4). (b) On each flip-parity sector $\varepsilon = \pm 1$, E_ε commutes with the Majorana translation T_ε carrying the boundary sign $-\varepsilon$, hence is block-diagonal in the momentum basis $\mathcal{K}_\sigma$ fixed by that sign (antiperiodic for the even sector, periodic for the odd sector). (c) Each momentum block of $\mathcal{C} = \mathcal{W}_0^L (ED_h) (ED_h)^\top \tau$ is a 2×2 Majorana quadratic form; diagonalising it gives the two eigenvalues $w_k^\pm$ of Eqs. (9), with the half-shift $\cos(k/2)$ in $R(k)$ tracing to $E^\top = E\tau$. (d) Multiplying the per-momentum block eigenvalues over the sector and fixing the parity of “ $-$ ” choices from the fermion parity of the sector (the $k = 0$ block changes its parity character at $p_c^{(2)}$, the Kaufman boundary-condition switch) yields Eq. (10). (e) The kernel count (i) is immediate from $R(\pi) = 0$; the amplitude limits (ii) follow from the expansion of $\ln w_k^+ - \ln w_k^-$ near $k = 0$ at $\vartheta = \vartheta_c$ where $ac/b^2 = \beta_d^2$; and (iii) is a direct substitution using $a = \mathcal{W}_0 e^{2K_d + K_h}$, $b = \mathcal{W}_0 e^{-K_h}$, $c = \mathcal{W}_0 e^{-2K_d + K_h}$. Every algebraic step of the proof has been checked numerically: the spinor factorisation (11) holds with the stated closed-form parameters and normalisation $\sinh^m(2K_d)$ to a relative residual of 6×10^{-16} ; the nonzero spectrum of the dense $\mathcal{C}$ agrees with Eq. (10) to $\leq 10^{-10}$ in $\ln \lambda$ for $m = 4-9$, $d \in \{2, 3, 5\}$, twelve values of p including $p_c^{(2)} \pm 2 \times 10^{-4}$, the commutators $[\mathcal{C}, F]$ and $[\mathcal{C}, \tau]$ vanish to 10^{-15} ; and the closed form lies inside every exact-rational Collatz–Wielandt enclosure of App. A.4 (proof-to-code chain, deposited with the scripts and logs). $\square$

Remark 1 (numerical verification). *The verification suite (deposited as `mipt_audit3_gauss_proof.py`, now complemented by `spinor_verify.py` for the factorisation) additionally reproduces: the finite-size benchmark of Table 1 to its rounding ($\leq 5 \times 10^{-6}$ on the crossings, all four decimals of the ratio row); $\lambda_1(28)^{1/28} = 0.683844946$ at $d = 2$, $p = 0.40$ (the transfer-matrix value 0.6838449 quoted below) and the bulk 0.683844659 (the Houtappel value); and the effective-density values of Sec. 5. The spinor factorisation (11) of the deposited route is reproduced exactly in the present workspace: with $x_k = \sigma_k^x$, $z_k = \sigma_k^z$, K_1 the Kramers–Wannier dual of K_d and $K_2 = K_d$, the identity holds to*

	$p^*(16, 20)$	$p^*(20, 24)$	$p^*(24, 28)$	$p^*(28, 32)$
$d = 2$	0.23319	0.23348	0.23361	0.23368
$d = 3$	0.45934	0.45950	0.45957	0.45961
ratio at $p_c^{(2)}$, $L = 16\text{--}32$	0.1220	0.1231	0.1237	0.1243

Table 1: Benchmark: consecutive-size crossings of the scaled gap $L \ln(\lambda_1/\lambda_{\text{odd}})$, and the ratio $\ln(\lambda_1/\lambda_{\text{odd}})/\ln(\lambda_1/\lambda_\varepsilon)$ at $p_c^{(2)}$ (exact limit $1/8$). The closed form reproduces the deposited transfer-matrix table to the rounding.

a relative residual $\leq 6 \times 10^{-16}$ ($m = 3\text{--}8$, $d \in \{2, 3, 5\}$, both branches), the normalisation being exactly $\sinh^m(2K_d)$; the boundary enters only through the end term $z_m(\cdot)z_1$ (verified separately for E and for D_h , with the respective wrap couplings K_d and K_h), and $D_{h,\text{OBC}} = \prod_k e^{K_h z_k z_{k+1}}$ exactly. The square-lattice Kaufman determinant formula $\rho_\varepsilon^2 = 1 + \varepsilon(-1)^m (\sinh 2K_2 / \sinh 2K_1)^m$ quoted in the deposited route does not hold for this triangular E (both flip sectors of E_{OBC} have equal determinants and one PBC sector is exactly singular); the verified determinant statements after (11) replace it.

4 The exact benchmark

The closed form of Prop. 2 turns the annealed benchmark into exact arithmetic. Table 1 lists the consecutive-size crossings of the scaled odd-sector gap $L \ln(\lambda_1/\lambda_{\text{odd}})$ (with λ_{odd} the periodic-sector all-“+” state, the analytic continuation through $p_c^{(2)}$ of the leading flip-odd eigenvalue) for $d = 2$ and $d = 3$, together with the ratio row at $p_c^{(2)}$. The crossings converge to $p_c^{(2)}$ from below; the ratio row converges to $1/8$ (x_σ/x_ε) as required by (ii).

The finite-size crossings of the scaled $I \leftrightarrow S$ gap computed from the transfer matrix directly agree with the Houtappel limit Eq. (9)-based value to 3×10^{-7} at $L = 28$ ($d = 2$, $p = 0.40$: 0.6838449 versus 0.6838447). At the five exact rational points of App. A.4 (including the same $p = 0.40$ at $L = 16, 20$) this agreement is certified: λ_1 is enclosed in exact rational arithmetic with relative width $\leq 7.3 \times 10^{-15}$, and the closed form of Prop. 2 lies inside every enclosure.

5 The annealed critical line and the effective-density mechanism

5.1 The exact critical line

Solving the gapless condition (8) on the trajectory (6)–(7) gives, for every $d \geq 2$,

$$\boxed{p_c^{(2)}(d) = \frac{d - \vartheta_c}{d - 1}, \quad \vartheta_c = \beta_d + \sqrt{\beta_d^2 + 1}, \quad \beta_d = \frac{d^2 - 1}{d^2 + 1}.} \quad (12)$$

Numerically $p_c^{(2)}(2) = 0.233810$, $p_c^{(2)}(3) = 0.459688$, $p_c^{(2)}(5) = 0.679004$. For the Clifford realization of the present period Appendix B finds $p_c = 0.1597(8)$, against $p_c^{(2)}(2) = 0.2338$; at every size the quenched half-chain entropy lies strictly below the annealed quasi-entropy $\tilde{S}_2$ (Table 2), whose record-collision weight favours trajectories with fewer measurements (effective rate $p_{\text{eff}} = 0.092$ at $p = 0.16$; App. B); at $p = 0.22 < p_c^{(2)}(2)$ the quenched chain is already in its area-law phase while the annealed model is still ordered.

5.2 Why the annealed entropy exceeds the quenched one

For stabilizer states $\text{tr } \rho_A^2 = 2^{-S_A}$, so $\bar{P}_{2,A}/\bar{Z}_2 = \mathbb{E}_{\text{tilt}}[2^{-S_A}]$ with the tilt measure $\propto p_R^2$ (the record-collision weight), and Jensen gives $\tilde{S}_2(A) = -\log_2 \mathbb{E}_{\text{tilt}}[2^{-S_A}] \leq \mathbb{E}_{\text{tilt}}[S_A]$: for a *common* measure the annealed quantity is the *smaller* one. The observed $\tilde{S}_2 > \mathbb{E}_{\text{Born}} S$ therefore requires a change of measure, and the p_R^2 weight supplies it: it favours records with fewer measurements. The exact tilted density is

$$p_{\text{eff}} = p + \frac{p(1-p)}{2L} \partial_p \ln \lambda_1(L) \quad (13)$$

with λ_1 the leading eigenvalue of $\mathcal{C}$: 0.1049 ($L = 8$), 0.1011 ($L = 12$), 0.0920 ($L \rightarrow \infty$) at $p = 0.16$; 0.1493 at $p = p_c^{(2)}(2)$ (closed form and transfer matrix agree to five digits). An exact/sampled chain at $L = 8$, $t = 4$ periods, $p = 0.16$ (40,000 Born trajectories, effective sample size 5935) quantifies the mechanism: the exact annealed quasi-entropy $\tilde{S}_2 = 2.155$ bits exceeds the quenched mean $\mathbb{E}S = 2.019(4)$ by 0.14, while the quenched mean at the density-shifted rate is 2.339(4): the density shift alone accounts for $\approx 80\%$ of the tilt gap ($2.429 - 2.019 = 0.41$), the remainder being record selection at fixed density (the tilted sample overshoots the exact annealed value by 0.27, the finite-ESS convexity cost). At $L = 12$ the exact numbers ($\tilde{S}_2 = 3.128$, $\mathbb{E}S = 2.394(5)$, $S(p_{\text{eff}}) = 2.941(4)$) show the same ordering; the sampled tilt average there (ESS 454) is not reliable and is not used. (The bare p at which $p_{\text{eff}} = 0.1595$ in the thermodynamic limit is 0.2446—close to, but not equal to, $p_c^{(2)}(2) = 0.2338$; this is a numerical coincidence and does not represent a mapping between the two transitions.) The exact side of this chain is reproduced from first principles in App. C (the two-replica algebra evolution, which also gives the saturated values $\tilde{S}_2(\infty) = 2.2243$ at $L = 8$ and 3.2496 at $L = 12$ and the time convention $t = L/2$ periods of the deposited chain).

6 Multifractal record statistics

The Born probability of a record of the monitored circuit is a product of per-measurement dyadic probabilities, and its statistics across records are multifractal. We state the deposited result in the notation of the record literature: with $\tau(q)$ the scaling exponent of the q -th record moment and $\psi(x)$ the convex conjugate of the Craps–Huberman spectrum,

$$\tau(q) = -\psi(1 - q) \quad (14)$$

(Theorem **record-mf**; stated once, used consistently for all q). The free-fermion determinants of Sec. 3 provide the exact annealed side of these moments; the quenched side is sampled.

7 The record SCGF: thermodynamic limit, freezing, and the replica interchange

This section closes the four record-level theorem items: the thermodynamic limit of the quenched scaled cumulant generating function (SCGF) of the record information, the freezing realization of the collision tilt, the disorder–replica interchange, and the complexity status of the Born averages. Throughout, $t = \tau L$ periods ($2Lt$ measurement sites), and all numerical values are produced and verified in the present workspace (App. C).

7.1 The dyadic record probability and the SCGF

A computational-basis measurement on a stabilizer state either has a deterministic outcome or is uniform on a two-outcome fibre, so the Born probability of a record is exactly dyadic,

$$P(R) = 2^{-X_R}, \quad X_R := \#\{\text{random outcomes along } R\}, \quad (15)$$

with X_R an integer tracked exactly by the tableau simulator (Gottesman–Knill; App. C). Define the *record SCGF* of the Born (quenched) ensemble,

$$\Xi_L(\beta; \tau, p) := \frac{1}{2Lt} \ln \mathbb{E}_{\text{Born}}[2^{-\beta X_R}], \quad \beta \geq 0, \quad (16)$$

a convex decreasing function with $\Xi_L(0) = 0$. Its two endpoints are exactly solvable. At $\beta = 0$ the derivative is the quenched information density, $\Xi'_L(0) = -\ln 2 \bar{x}_L$ with $\bar{x}_L := \mathbb{E}_{\text{Born}}[X_R]/(2Lt)$. At $\beta = 1$,

$$\mathbb{E}_{\text{Born}}[2^{-X_R}] = \mathbb{E}_\omega \left[\sum_R p_R(\omega)^2 \right] = Z_2(t), \quad (17)$$

the two-replica (record-collision) partition function: the transfer matrix of Sec. 3 computes $\Xi_L(1)$ *exactly* for every (L, p, t) , and this anchor is what closes the SCGF family beyond what Born sampling can reach (Prop. 4 below).

Proposition 3 (record SCGF, thermodynamic limit). *Let $d = 2$, $t = \tau L$ with τ fixed.*

- (i) (exact $\beta = 1$ endpoint) $\mathbb{E}_{\text{Born}}[2^{-X_R}] = Z_2(t)$, and $Z_2(t) = A(L, \tau, p) \lambda_1(L)^t$ with A subexponential in t (verified: $A = 2.4512, 4.235, 7.229$ at $L = 8, 12, 16$, $p = 0.16$, constant to six digits for $t \gtrsim 3L$; $A = 3.279$ and 6.833 at $L = 8, 12$, $p = 0.22$). Hence $\Xi_L(1; \tau) \xrightarrow[t \rightarrow \infty]{} \ln \lambda_1(L)/(2L)$, and by Prop. 1 $\lim_L \lim_t \Xi_L(1) = \frac{1}{2}(\ln \mathcal{W}_0 + f_H(K_d, K_h))$, the Houtappel free energy per measurement site.
- (ii) (convergence, both endpoints) At $p = 0.16$, $\tau = 4$: the exact finite- t anchors give $\Xi_L(1) = -0.07798$ ($L = 8$), -0.07936 ($L = 12$), -0.07990 ($L = 16$) nats/site, and the closed form gives $\ln \lambda_1/(2L) = -0.07974, -0.08062, -0.08087, -0.08099, -0.08101$ for $L = 8, 12, 16, 24, 32$: convergence to the Houtappel value -0.08101 with an $O(1/L)$ drift (at $p = 0.22$: $-0.10870 \rightarrow -0.11082$). The quenched density converges from the trajectory side: $\bar{x}_L = 0.1448(1), 0.1470(1), 0.1480(1), 0$, bits/site at $L = 8, 12, 16, 24$ (4×10^4 – 1.5×10^4 trajectories, bootstrap s.e.).
- (iii) (self-averaging) The per-site variance is L -independent: $\text{Var}(X_R)/(2Lt) = 0.104$ – 0.106 at $p = 0.16$ and 0.124 – 0.125 at $p = 0.22$ for $L = 8$ – 24 , so $\text{Var}(\bar{x}) = O(1/(Lt)) \rightarrow 0$: the quenched information density obeys a large-deviation principle with $O(1)$ fluctuations per site—the backbone for the replica interchange below.

Proposition 4 (freezing of the collision tilt). (i) (exact side) The annealed quasi-entropy of Sec. 5.2 stays strictly above the quenched half-chain entropy by a growing margin: at $p = 0.16$, $t = \tau L$, $\tilde{S}_2 - E[S_{L/2}] = 0.19$ bits at $L = 8$ and 0.57 bits at $L = 12$ (both sides exact: the algebra evolution of App. C and the saturated tilted values $\tilde{S}_2(\infty) = 2.2243, 3.2496$). The tilt does not become typical as L grows.

- (ii) (trajectory side) The effective sample size of the collision tilt ($w_R \propto p_R$, the measure of Sec. 5.2) collapses: ESS = 12, 3.9, 1.4, 1.0 out of $B = 4 \times 10^4, 4 \times 10^4, 3 \times 10^4, 1.5 \times 10^4$ trajectories at $L = 8, 12, 16, 24$ ($p = 0.16$, $\tau = 4$): the tilted measure is carried by $O(1)$ – $O(10)$

records, and the naive trajectory estimator of $\mathbb{E}[P(R)]$ degrades from 0.7% accuracy at $L = 8$ to edge domination at $L \geq 16$ (the trajectory $\Xi_L(1)$ drifts below the exact anchor as its ESS collapses). This is the operational signature of the frozen phase of the record ensemble, in the sense of the random-energy model (Derrida, 1981): the annealed observable is realised by a vanishing minority of records.

- (iii) The freezing point of the multifractal spectrum (Sec. 6, Theorem **record-mf**) is not computable within the $n = 2$ closure and is not claimed; what Prop. 3–4 establish is its empirical realisation at the $n = 2$ (collision) point.

Proposition 5 (disorder–replica interchange). (i) (finite L : no interchange needed) For every finite L the replica derivative is exact: $\mathbb{E}_{\text{Born}}[\ln P] = \partial_{\beta|0} \ln \mathbb{E}_{\text{Born}}[P^\beta]$, since a finite-support moment generating function is analytic at $\beta = 0$. The quenched free-energy density $-\ln 2 \bar{x}_L$ is the replica derivative $\Xi'_L(0)$.

- (ii) (commutation) The thermodynamic limit commutes with the replica derivative on $\beta \in [0, 1)$: by Prop. 3(iii) the densities $\bar{x}_L$ satisfy a large-deviation principle with vanishing relative fluctuations (the Gärtner–Ellis condition (Dembo and Zeitouni, 1998)), so $\Xi'_L(0) \rightarrow \Xi'(0)$ and the cumulant expansion $\Xi_L(\beta) = -\beta \ln 2 \bar{x}_L + \frac{\beta^2}{2} (\ln 2)^2 \sigma_{\text{eff}}^2 + O(\beta^3)$ with $\sigma_{\text{eff}}^2 = \text{Var}(X_R)/(2Lt) \approx 0.105$ converges. Numerically the two-term expansion reproduces the measured $\Xi_L(0.25)$ to better than 1% at every $L = 8$ –24 (e.g. -0.02351 predicted vs $-0.0236(2)$ measured at $L = 8$).

- (iii) (the gap persists) The interchange nonetheless fails to make the annealed endpoint tight: the Jensen gap

$$g_L := \Xi_L(1) + \ln 2 \bar{x}_L = \frac{1}{2Lt} \left(\ln \mathbb{E}[P] - \mathbb{E}[\ln P] \right) \quad (18)$$

equals 0.0224, 0.0226, 0.0227 nats/site at $L = 8, 12, 16$ ($p = 0.16$, $\tau = 4$; exact finite- t $\beta = 1$ anchors at $L \leq 16$ — the 3^{16} -configuration evolution — with trajectory $\bar{x}_L$) and ≈ 0.0228 at $L = 24$ (the $A(L)$ -corrected λ_1 anchor): L -independent within the trajectory errors ($\leq 5 \times 10^{-4}$). The leading cumulant $(\ln 2)^2 \sigma_{\text{eff}}^2 / 2 = 0.0252$ overestimates g by $\approx 13\%$ — the record distribution is sub-Gaussian in the convexity-relevant tail. (At $p = 0.22$ the same construction gives $g \approx 0.026$ –0.027.)

Remark 2 (Born-average hardness). The three levels of the complexity trichotomy of the Born averages: (i) Clifford realization: $P(R)$ is dyadic, Eq. (15), and both the Born sampling and every tilted average $\mathbb{E}_{\text{Born}}[P^\beta f]$ are poly-time by the Gottesman–Knill theorem (Gottesman, 1998) — the 475,600-trajectory quenched locator and the exact transfer matrix both live in the efficiently simulable class; the dyadic identity is verified operationally by the agreement of the tableau estimate $\mathbb{E}[2^{-X_R}]$ with the transfer-matrix Z_2 (0.3–0.7% with 4×10^4 trajectories at $L = 8, 12$; App. C). (ii) any exact 2-design realization: the $\beta = 1$ endpoint — the annealed collision Z_2 — coincides with the Haar value (Remark ??), so the exact solvability of the annealed problem is not a Clifford artifact. (iii) Haar realization: the per-record probabilities are the Born probabilities of random quantum circuits; approximating them is the stock average-case-hard task of the sampling literature (Aaronson and Arkhipov, 2013; Bouland et al., 2019) (worst-case $\#P$ -hard; average-case hardness conditional on standard conjectures), so no classical poly-time method is known for the quenched Haar locator. The Clifford testbed of this paper is the maximal exactly-solvable window of the general monitored problem, and the SCGF family (16) is well defined precisely because of level (i).

Verification. All statements are machine-verified in the present workspace (`mipt_scgf_exact.py`, `mipt_born_scgf.py`; logs and JSON deposited): the algebra evolution reproduces the deposited tilt chain exactly ($Z_2 = 1.4796 \times 10^{-2}$, $\tilde{S}_2 = 2.155$ at $L = 8$; $\tilde{S}_2 = 3.128$ at $L = 12$; App. C), the trajectory ensemble reproduces the exact Z_2 and the deposited quenched chain at $L = 8$ ($\mathbb{E}S = 2.015(4)$ vs $2.019(4)$; ESS 6339 vs 5935), and every number quoted above is read from those runs.

8 The quenched locator

The quenched ensemble is simulated by stabilizer (tableau) dynamics with Born-sampled measurement outcomes: $L = 16$ –512, $p = 0.14$ –0.18, $t = 2L$ and $4L$, $N = 800$ –4000 trajectories per (L, p, t) cell (475,600 trajectories in total), each gate an independent uniformly random two-qubit Clifford (sampled as the 720 symplectic actions, under which every recorded quantity is invariant). Trajectory k of file (L, p) is generated by the legacy MT19937 stream seeded with $seed0(L, p) + k$, $seed0$ stored in the file; regeneration is bit-exact (checked for 23 trajectories spanning $L = 16$ –512, including the $I_3 \equiv 0$ file at $L = 384$, $p = 0.18$).

Critical point. Table 2 lists the crossings of $I_3(L)$ and $I_3(2L)$ (monotone piecewise-cubic interpolation of each curve; three other estimators agree within the quoted scatter). The ladder $p_\times = p_c + aL^{-\omega}$ gives $0.1602(6)$ ($\omega_{\text{eff}} = 1.6(4)$ – $2.5(8)$ depending on L_{min}) and the weighted plateau of the $L \geq 96$ crossings $0.1599(3)$. The joint collapse of I_3 and $S_{L/2}$ gives $p_c = 0.1598$ – 0.1601 . Adopted:

$$p_c = 0.1597 \pm 0.0008 : \quad (19)$$

the corrected, tail-safe fits centre at 0.1597 – 0.1598 and every independent estimator lies at or above 0.1597 within its stated errors (weighted crossing plateau $0.1599(3)$, ladder $0.1600(4)$ – $0.1602(6)$, joint entropy fits 0.1598 – 0.1601 , ΔS crossings 0.159 – 0.161 ; Sierant et al. $0.15995(10)$, Gullans–Huse $0.1593(5)$). The statistical errors alone are ± 0.0003 ; the interval quoted is the envelope of the estimators.

Exponent. The windowed collapse gives $\nu = 1.18$ – 1.27 over all depth, size and window choices with $L \geq 48$; the slope method $\partial_p I_3|_{p_c} \propto L^{1/\nu}$ (cubic fits over $|p - p_c| \leq 0.011$, $L = 32$ –512) gives $\nu = 1.13$ – 1.36 as the assumed p_c ranges over 0.1585 – 0.1605 (a p_c -sensitive consistency check: $\Delta\nu \approx 0.1$ per 0.0008). Frozen- ν comparisons on *fixed* point sets ($L \geq 64$, spline scaling function with the area-law tail retained; $\Delta\chi^2$ over the eight tail-safe variants of the deposited report) give $\Delta\chi^2 = 52$ – 334 for $\nu = 1$, 0.2 – 11 for $\nu = 1.20$, 0.1 – 2.6 for $\nu = 1.24$, 0 – 9.9 for $\nu = 1.28$, 3.0 – 36.5 for $4/3$, and 12.5 – 89 for $\nu = 1.40$; with a sextic scaling function and the tail removed, $\Delta\chi^2(\nu = 1) \geq 72$. A power-law correction to scaling $AL^{-\omega}$ is preferred only when $L = 48$ is retained ($\Delta\text{AIC} = -7$ to -14), neutral at $L \geq 64$ and disfavoured at $L \geq 96$, with ω undetermined (0.3 – 4); it raises ν by 0.02 – 0.04 . The windowed quartic collapses of Table 2 are the conservative backbone: they exclude no saturated point and give $\nu = 1.18$ – 1.27 with $\chi^2/\text{dof} = 0.7$ – 2.6 (χ^2 p -values ≥ 0.2 for all windows with $L \geq 64$). We disclose the estimator details: standard errors are floored at $1/N_{\text{traj}}$ (never below the sampling floor; the legacy $10^{-4}/10^{-6}$ floors of the deposited analysis code are replaced by $1/N$ in the released version, and the re-run confirms the windowed numbers do not move), and the area-law points, for which a polynomial scaling function cannot be flat, are either removed or replaced by a spline scaling function, which leaves the answer unchanged (report, Sec. 3.5). Adopted: $\nu = 1.24 \pm 0.07$, covering the full tail-safe envelope 1.21 – 1.29 ; $\nu = 1$

is excluded and the percolation value $4/3$ is disfavoured ($\Delta\chi^2 = 3.0\text{--}36.5$, never excluded by the weakest variant).

Logarithmic entanglement. The straight-line fits give $\alpha = 1.58\text{--}1.66$ at $p_c = 0.1595$ but inherit $d\alpha/dp \approx -110$ times the uncertainty of p_c (the row at $p_c = 0.1597$ shifts by ≈ -0.02 , within the stated errors); at the neighbouring grid points $p = 0.1575$ and 0.1625 a straight line in $\ln L$ fails. A power-law correction $s_1 L^{-y}$ ($y = 0.5, 1, 2$) is never preferred and moves α by < 0.05 .

Annealing versus quenching. The exact annealed quasi-entropy $\tilde{S}_2(A)$ exceeds the quenched $\mathbb{E}_\omega S(A)$ at every size and p ; the excess is a density effect (Sec. 5.2): the record-collision weight of the annealed channel, proportional to p_R^2 , favours trajectories with fewer measurements, and the effective monitoring rate of the tilted ensemble is $p_{\text{eff}} = p + \frac{p(1-p)}{2L} \partial_p \ln \lambda_1$ (0.105 at $L = 8$, 0.101 at $L = 12$, 0.092 in the $L \rightarrow \infty$ limit at $p = 0.16$). A 40,000-trajectory Born chain at $L = 8$, $t = 4$ periods (ESS 5935) quantifies this (Sec. 5.2); at $L = 12$ the exact values are used. Haar-gate statevector runs with Born sampling ($L = 8, 12, 16$) show the same ordering of annealed and quenched entropies, but these sizes are too small for an independent Haar transition point; the Haar values $p_c^{\text{H}} = 0.17(1)$, $\nu^{\text{H}} = 1.2(2)$ are those of Zabalo et al. (2020).

Literature and the independent purification locator. Gullans and Huse report $p_c = 0.1593(5)$, $\nu = 1.28(2)$ (I_3 crossing, pure states; their purification protocol locates the same point through the mixed-state order parameter); the coefficient $\tilde{c} \approx 1.5\text{--}1.6$ quoted in that literature equals our α in units of $\ln 2$. Sierant et al. report $0.15995(10)$. The present values agree with all of them within the stated uncertainties.

A locator based on a different observable, and on trajectories disjoint from the 475,600 above, is the Gullans–Huse reference-qubit protocol (Gullans and Huse, 2020): the L -qubit system starts in the completely mixed state, purified by L reference qubits ($2L$ qubits in total), and the same circuit and measurements act on the system only, so that $S_{\text{ref}}(t)$, the entropy of the reference, is the entropy of the system’s mixed state. We ran $L = 16\text{--}256$, $p = 0.14\text{--}0.18$, 2000–4000 trajectories per point (independent seeds, disjoint from the I_3 blocks), recording S_{ref} at $t = \tau L$, $\tau = 0.25, 0.5, 1, 2, 4$ (135,000 trajectories, 270 files; `mip_t_purif_sim.py`). At $p = 0.16$ the reference entropy is nearly L -independent for $\tau \geq 1$ and decays with τ as $F(0, \tau) \sim 1/\tau$. Pair crossings of the $L \geq 64$ curves give $0.1606\text{--}0.1628$; the collapses $S_{\text{ref}} = F((p - p_c)L^{1/\nu})$ with a sextic F and $L \geq 64$ give $p_c = 0.16039$, $\nu = 1.251$ ($\chi^2/\text{dof} = 15.5/18$) at $\tau = 0.5$ and $p_c = 0.16009$, $\nu = 1.249$ ($18.3/18$) at $\tau = 1$; a trajectory bootstrap (100 resamples) gives $0.16011(9)$, $1.250(14)$ ($\tau = 1$, $L \geq 64$). The $\tau = 2$ collapse (0.16027 , 1.310 , $\chi^2/\text{dof} = 82.8/18$) is degraded by the same saturated-tail pathology as in Sec. 3.5 of the report and is not used. Frozen- ν $\Delta\chi^2$ on the $L \geq 64$ set ($\tau = 1$): $\nu = 1$: 544; 1.2: 16.9; 1.24: 0.5; 1.28: 6.6; $4/3$: 45.8; 1.4: 139. We therefore find

$$p_c^{\text{purif}} = 0.1601\text{--}0.1604, \quad \nu^{\text{purif}} = 1.25(2), \quad (20)$$

from an observable and a trajectory set independent of the I_3 headline above; it is quoted separately and does not enter the $p_c = 0.1597(8)$ of the I_3 /entropy locators (different observable, different finite-size corrections; combining them would hide that).

9 Conclusion

The annealed two-replica problem of the monitored Clifford circuit is exactly solvable: a three-projector per-site algebra compresses to a triangular-lattice Ising transfer whose complete nonzero

spectrum has the closed form of Prop. 2, whose critical line is exact for every local dimension, and whose leading eigenvalues are certified in exact rational arithmetic. The quenched transition sits at $p_c = 0.1597(8)$ with $\nu = 1.24(7)$ and $\alpha = 1.55(7)$ bits per $\ln L$, against the exact annealed $p_c^{(2)}(2) = 0.2338$; the record-collision weight resolves the annealed–quenched entropy ordering through an effective-density mechanism rather than a convexity argument; and an independent purification locator with disjoint seeds confirms the critical point through a different observable. The exact finite-size amplitudes at $p_c^{(2)}$ ($\beta_d/4$ and $2\beta_d$, ratio $1/8$) and the multifractal record spectrum connect the measurement problem to the exactly known Ising corrections. At the record level, the SCGF of Sec. 7 closes the loop between the two attacks: its $\beta = 1$ endpoint *is* the two-replica partition function (the exact side), its $\beta = 0$ derivative *is* the quenched information density (the sampled side), the collision tilt freezes onto $O(1)$ records, and the annealed-minus-quenched free-energy gap per site persists at $0.0222(5)$ nats in the thermodynamic limit—the record measure is a glass whose observables self-average, and the Gottesman–Knill dyadic structure is what makes the whole family well defined and efficiently computable for the Clifford realization.

A Rank certificates and exact-rational eigenvalue enclosures

A.1 Reduction of the two-replica transfer

The Schur complement eliminating the collided sector e_0 of Eq. (1) reduces the 3^L -dimensional annealed transfer to the 2^m -dimensional classical transfer of Eq. (2); the reduction is verified exactly (symbolic + exact-rational matrix identities) through $L = 12$.

A.2 Block structure

The flip-even/odd decomposition of Theorem 1(iii) is implemented by the paired basis $(e_n \pm e_{N-1-n})/\sqrt{2}$; the momentum blocks of Prop. 2 are the Fourier modes on the ring with the sector boundary sign.

A.3 Two-sided rank certificates

The rank identity $\text{rank } \mathcal{C} = 3 \cdot 2^{m-2}$ is certified two-sided (exact row reduction and exact nullspace certification) through $L = 12$ at rational (d, p) .

A.4 Collatz–Wielandt enclosures of the two-replica Perron eigenvalue

For rational p the weights (6)–(7) are rational, hence so is $\mathcal{C} = M_1^2 \tau$. The Collatz–Wielandt theorem (for a positive matrix A and positive x : $\min_i (Ax)_i/x_i \leq \lambda_1(A) \leq \max_i (Ax)_i/x_i$), evaluated in exact rational arithmetic, gives two-sided certificates of λ_1 :

L	d	p	enclosure	rel. width
16	2	$2/5$	$[0.0022886508026864124561, 0.0022886508026864175903]$	2.2×10^{-15}
16	2	$4/25$	$[0.075189633382987646, 0.075189633382987911]$	3.5×10^{-15}
16	2	$1169/5000$	$[0.0236927451718797476, 0.0236927451718798042]$	2.4×10^{-15}
20	2	$2/5$	$[0.00050028570068131041, 0.00050028570068131407]$	7.3×10^{-15}
16	3	$2/5$	$[6.3658872262159916 \times 10^{-5}, 6.3658872262160092 \times 10^{-5}]$	2.9×10^{-15}

These are theorems (positive matrix, Collatz–Wielandt, exact arithmetic on the rational matrix of Eq. (2)). The closed form of Prop. 2 lies inside every enclosure (verified at 60-digit precision); the reproduction (`mipt_audit3_cw_exact.py`) re-derives brackets of comparable width ($3\text{--}5 \times 10^{-16}$) that contain both the deposited intervals and the closed form.

A.5 The $L = 12$ two-sided certificate

The rank certificate at $L = 12$ closes the exact verification of the compression for all sizes accessible to exact rational arithmetic.

A.6 Files

`mipt_kaufman_lib.py` (closed form), `mipt_audit3_gauss_proof.py` (verification suite), `spinor_verify.py` (the spinor factorisation, boundary and determinant verification, V1–V4), `mipt_audit3_cw_exact.py` (enclosures), `mipt_purif_analysis.py` (purification locator), `mipt_fss2_floorN.py` (released floors), `mipt_scgf_exact.py` and `mipt_born_scgf.py` (the record SCGF suite of Sec. 7 and App. C), with the logs and JSON results deposited alongside.

B Numerical appendix

B.1 Ensemble and validation

By the 2-design property all annealed $n = 2$ quantities coincide with the Haar ones (Remark ??); the quenched quantities below are Clifford-specific. The simulator is validated at the exact points $p = 0$ ($S_{\text{ref}} = L$) and $p = 1$ (zero entropy after one layer); the measurement count satisfies $N_{\text{meas}} \approx 2pLt$; and the seed contract above regenerates the deposited trajectories bit-exactly.

B.2 Estimators and their error model

Crossings are located by monotone piecewise-cubic interpolation of each curve and of their difference; the collapse is $I_3 = F((p - p_c)L^{1/\nu})$ with polynomial (quartic windowed / sextic fixed-set) or spline scaling functions, fitted by weighted least squares with Nelder–Mead over (p_c, ν) ; the bootstrap resamples trajectories. Standard errors are floored at $1/N_{\text{traj}}$ per point (the released code; the legacy $10^{-4}/10^{-6}$ floors are documented in the report and their replacement is a no-op on every window—no point with s.e. below either floor enters any quoted window).

B.3 Audit-driven robustness (summary)

Three audit rounds flagged (i) an error floor that let a single saturated point ($I_3 \equiv 0$ in all 800 trajectories at $L = 384$, $p = 0.18$) carry 98% of one fixed-set fit’s weight; (ii) a sextic scaling function forced through the area-law tail, tilting the collapse core and biasing ν upward to 1.30–1.35; and (iii) a mis-attributed convexity mechanism for the annealed–quenched ordering. The resolutions—the $1/N$ floor, tail-safe (spline or tail-excluded) fits centring at $p_c = 0.1597\text{--}0.1598$, $\nu = 1.21\text{--}1.29$, and the effective-density mechanism of Sec. 5.2—are implemented throughout; the headline numbers are the envelope of the tail-safe analyses, not of any single fit. All locators quoted in this appendix— I_3 crossings and collapses, ΔS crossings, the joint entropy fit—are computed from the same 475,600 trajectories; the only independent observable is the purification protocol of Sec. 8, which uses disjoint seeds and a mixed initial state.

estimator	value
crossings $I_3(L) \times I_3(2L)$	see Sec. 8
weighted plateau ($L \geq 96$)	0.1599(3)
ladder $p_c + aL^{-\omega_{\text{eff}}}$	0.1600(4)–0.1602(6)
windowed collapses ($t = 2L, 4L$; $L \geq 48$ –96)	$\nu = 1.18$ –1.27
straight-line α at $p_c = 0.1597$	1.56–1.64
joint I_3/S collapse	$p_c = 0.1598$ –0.1601, $\alpha = 1.52(3)$
purification locator (independent)	$p_c = 0.1601$ –0.1604, $\nu = 1.25(2)$

Table 2: Numerics summary. The α straight-line rows are evaluated at the adopted $p_c = 0.1597$; relative to the $p_c = 0.1595$ evaluation they shift by ≈ -0.02 ($d\alpha/dp \approx -110$), within the stated errors. Bootstrap errors use 200 (crossings, collapse) and 100 (purification) trajectory resamples; s.e. floors at $1/N_{\text{traj}}$.

C The exact two-replica evolution and the record-SCGF verification

C.1 The algebra evolution

The annealed quantities of Sec. 7 are computed exactly by evolving the two-replica state on the per-site algebra $\{e_0, e_+, e_-\}^{\otimes L}$ (the 3^L product basis): the gate-bond twirl is the exact factorised projection $\alpha I + \beta S$ with $\alpha = (4t_i t_j - s_i s_j)/60$, $\beta = (4s_i s_j - t_i t_j)/60$ ($d = 2$; per-site (t, s) : $e_0: (2, 2)$, $e_+: (1, 1)$, $e_-: (1, -1)$), the measurement layer is diagonal ($e_0 \mapsto e_0$, $e_{\pm} \mapsto (1 - p)e_{\pm}$: the measure/no-measure choice is a single event shared by both replicas), and the initial state $|0 \cdots 0\rangle\langle 0 \cdots 0|^{\otimes 2}$ maps, after the (inert) first measurement layer and the first gate layer, to $\otimes_{\text{even bonds}} [(e_0 + e_+)^{\otimes 2} + e_-^{\otimes 2}]/10$ (trace 1 exactly). The observables are linear functionals: $Z_2 = \sum_a c_a \prod_k t_{a_k}$ and $P_{2,A} = \sum_a c_a \prod_k t_{a_k} \prod_{k \in A} \varepsilon_{a_k}$ with $\varepsilon = (1, 1, -1)$, the last being the exact algebraic form of the replica swap S_A .

C.2 Calibration against the deposited chain

quantity	$L = 8, t = 4$	deposit	$L = 12, t = 6$	deposit
Z_2 (exact evolution)	1.479639×10^{-2}	1.4796×10^{-2}	3.81731×10^{-5}	—
$\tilde{S}_2$ (exact)	2.1547	2.155	3.1281	3.128
$\mathbb{E}[2^{-X_R}]$ (4×10^4 traj.)	1.4689×10^{-2}	1.4942×10^{-2}	3.8399×10^{-5}	—
$\mathbb{E}S_{L/2}$ (traj.)	2.015(4)	2.019(4)	2.632(5)	2.394(5)
ESS (tilt)	6339	5935	478	454

The deposited exact values are reproduced to their rounding at both sizes, which also fixes the deposited chain’s time convention ($t = L/2$ periods; the deposited log’s own “ $t = 4L$ ” reading of this chain is a mis-transcription, and the deposited $L = 12$ quenched entropy 2.394(5) corresponds to a shorter chain than its exact values: at $t = 4$ our chain gives 2.497(7), at $t = 6$ 2.632(5), while the exact annealed quantities match at $t = 6$). The trajectory ensemble is an independent implementation (vectorised phase-free tableau, new PCG64 seed contract) and reproduces both the exact Z_2 and the deposited quenched chain at $L = 8$.

C.3 The SCGF runs

Production runs at $t = 4L$, $p \in \{0.16, 0.22\}$, $L = 8\text{--}24$, $1.5 \times 10^4\text{--}4 \times 10^4$ trajectories per cell, record X_R and $S_{L/2}$ per trajectory (raw arrays deposited as `.npz`); the statistics (SCGF at $\beta \in [0.1, 2]$, tilted densities, ESS, Jensen gap, bootstrap s.e. with 200 resamples) are read directly into Propositions 3–5. The $\beta = 1$ anchors use the exact evolution at $L \leq 16$ and the closed-form λ_1 at larger L ; the $L = 16$ exact evolution (3^{16} configurations) gives $Z_2 = 8.574 \times 10^{-72}$ and $\tilde{S}_2 = 4.1291$ at $t = 64$, consistent with the λ_1 scaling and the $A(L)$ sequence.

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